SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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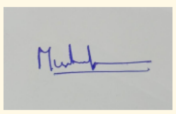
Section / Batch:		**Date:**	

Code of Conduct

I SYED MUHAFIZ A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

Annexure A – Class Test

1. Inflectional Morphology and Porter Stemmer

Input: *talked, talking, talks*

Morphological Analysis

Word	Root	Suffix Removed	Type	Porter Stemmer Rule	Normalized Root
talked	talk	**-ed**	Inflectional	Remove **-ed**	talk
talking	talk	**-ing**	Inflectional	Remove **-ing**	talk
talks	talk	**-s**	Inflectional	Remove **-s**	talk

Role of Morphological Normalization

- Removes grammatical suffixes to obtain the base form.
- Maps all variants to the common root **talk**.
- Reduces duplicate entries in the index.
- Improves information retrieval by matching all grammatical forms of the same word.

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2. Hybrid Morphological Processing Pipeline

Input: *relational, conditional, happiness, running*

Morphological Breakdown

Word	Root	Affix(es)	Type	FST Transition	Porter Stemmer Steps	Final Stem
relational	relate	-ional	Derivational	relate → +ional → relational	relational → relate → relat	**relat**
conditional	condition	-al	Derivational	condition → +al → conditional	conditional → condition → condit	**condit**
happiness	happy	-ness	Derivational	happy → +ness → happiness	happiness → happi	**happi**
running	run	-ing	Inflectional	run → +ing → running	running → run	**run**

Finite-State Morphological Parsing

```
Start
 |
 ├── relate ──(+ional)──▶ relational
 ├── condition ──(+al)──▶ conditional
```

```
├── happy ──(+ness)──▶ happiness
└── run ──(+ing)──▶ running
```

Contribution of Each Technique

- **Derivational analysis:** Identifies suffixes that create new words or change word class.
- **Inflectional analysis:** Removes grammatical endings while preserving meaning.
- **Finite-State Parsing:** Validates legal morphological transitions.
- **Porter Stemmer:** Produces a consistent stem for indexing and retrieval.

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3. Morphology-Driven SEO Solution

Input: *optimize, optimization, optimized, optimizing*

Morphological Analysis

Word	Root	Affix(es)	Type	FST Transition	Porter Stem	Normalized Form
optimize	optimize	—	Base Form	optimize	optim	optim
optimization	optimize	-ation	Derivational	optimize → +ation	optim	optim
optimized	optimize	-ed	Inflectional	optimize → +ed	optim	optim
optimizing	optimize	-ing	Inflectional	optimize → +ing	optim	optim

Finite-State Parsing

```
Start
 |
optimize
├── +ation ─▶ optimization
├── +ed ────▶ optimized
└── +ing ───▶ optimizing
```

Benefits

- Groups all keyword variants into one indexed term.
- Eliminates redundant entries in the search index.
- Improves search relevance and ranking.
- Ensures consistent retrieval of all related word forms.

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4. Hybrid Morphological Pipeline

Input: *compute, computer, computing, computation*

Morphological Breakdown

Word	Root	Affix	Type	FST Transition	Porter Stem	Final Stem
compute	compute	—	Base Form	compute	comput	**comput**
computer	compute	-er	Derivational	compute → +er	comput	**comput**
computing	compute	-ing	Inflectional	compute → +ing	comput	**comput**
computation	compute	-ation	Derivational	compute → +ation	comput	**comput**

State Transitions

```
Start
 |
compute
├── +er ──────▶ computer
├── +ing ─────▶ computing
└── +ation ─▶ computation
```

Normalized Representation

Word	Morphological Structure	Final Stem
compute	compute	comput
computer	compute + er	comput
computing	compute + ing	comput
computation	compute + ation	comput

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5. Morphology-Based Framework

Input: *decide, decision, decisive, deciding*

Morphological Breakdown

Word	Root	Affix(es)	Type	Meaning Change	Porter Stem	Final Stem
decide	decide	—	Base Form	Verb	decis	**decis**
decision	decide	-sion	Derivational	Verb → Noun	decis	**decis**
decisive	decide	-ive	Derivational	Verb → Adjective	decis	**decis**
deciding	decide	-ing	Inflectional	Present participle	decis	**decis**

Finite-State Parsing

```
Start
 |
```

```
decide
├── +sion ─▶ decision
├── +ive ──▶ decisive
└── +ing ──▶ deciding
```

Normalized Output

Word	Morphological Structure	Final Stem
decide	decide	decis
decision	decide + sion	decis
decisive	decide + ive	decis
deciding	decide + ing	decis

Framework Summary

- **Derivational rules** identify suffixes that change meaning or word class.
- **Inflectional transformations** remove grammatical endings without changing meaning.
- **Finite-state parsing** validates valid morphological transformations.
- **Porter Stemmer** reduces all variants to the common stem **decis**, improving semantic clustering and efficient information retrieval.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

Q. No. / Criteria Level	Understanding of Task	Procedure & Execution	Observation & Result	Viva Voce / Conceptual Response	Time Management & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________